

Module 14 - chapter 7

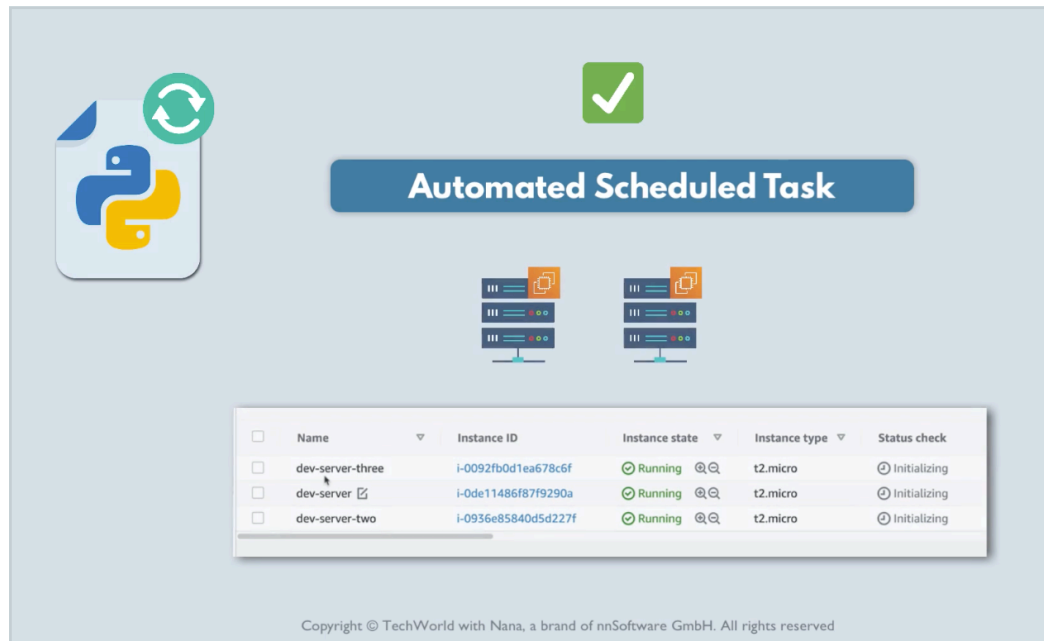
Configure Server: Add Environment Tags to EC2 Instances

The file for the project used in this lecture can be found here:

- <https://gitlab.com/twn-devops-bootcamp/latest/14-automation-with-python/automation-projects/-/blob/main/add-env-tags.py>

Project intro

We created our first automated task in python



Let's say we have a different scenario:

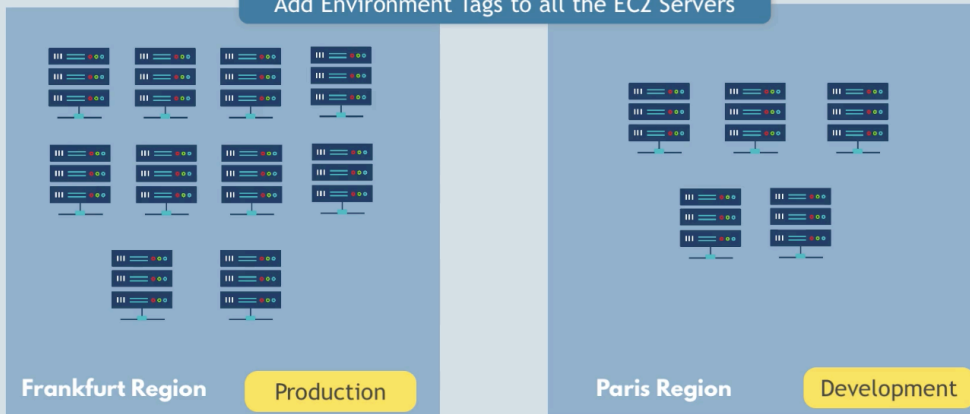
We have 20 servers in the Frankfurt region which are used as production servers

We also have 10 servers in the Paris region that we use as development servers.

And we created all those servers using TF and now we want to add environment tag all the servers

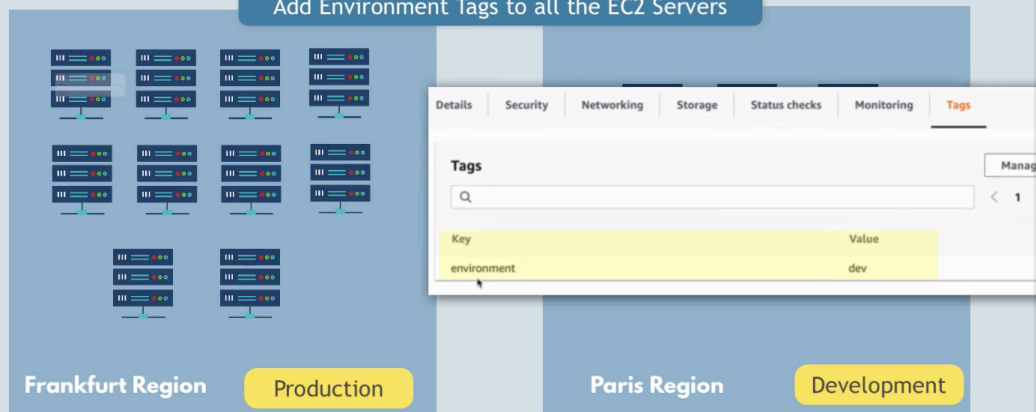
Project Exercise: Add Tags to EC2 Instances

Add Environment Tags to all the EC2 Servers



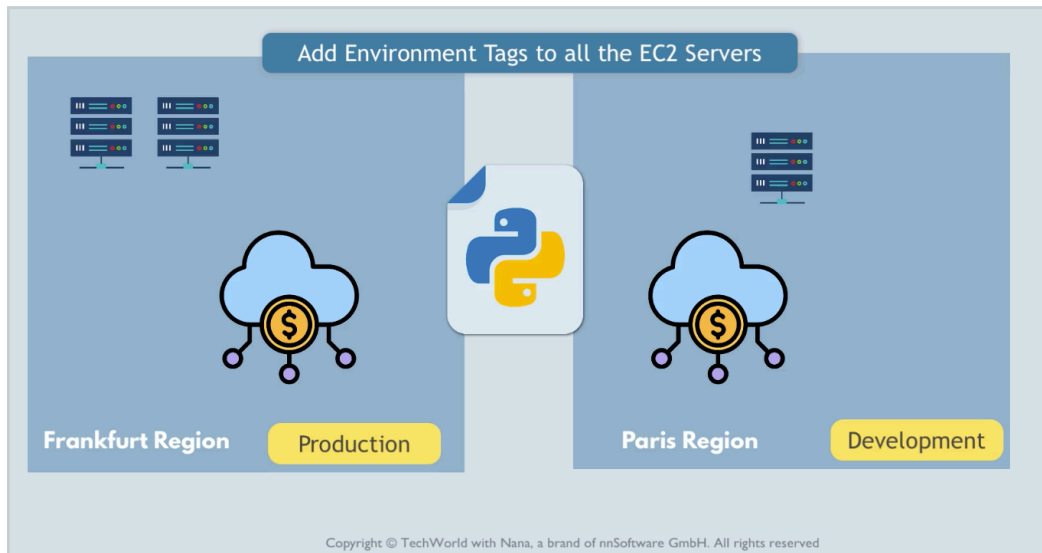
Copyright © TechWorld with Nana, a brand of nnSoftware GmbH. All rights reserved

Add Environment Tags to all the EC2 Servers



Copyright © TechWorld with Nana, a brand of nnSoftware GmbH. All rights reserved

And doing it manually is not efficient so we will use python, and for this demo for simplicity and cost we will create 2 instances for Frankfurt and 1 for Paris



Implementation

- 1 - Get all instances from a region
- 2 - Add tags to the instances

So let's get the instances from a region:

```
add-env-tags.py x
1 import boto3
2
3 ec2_client_frankfurt = boto3.client(*args: 'ec2', region_name='eu-central-1')
4
5
6 reservations_frankfurt = ec2_client_frankfurt.describe_instances()['Reservations']
7 for res in reservations_frankfurt:
8     instances = res['Instances']
9     for ins in instances:
```

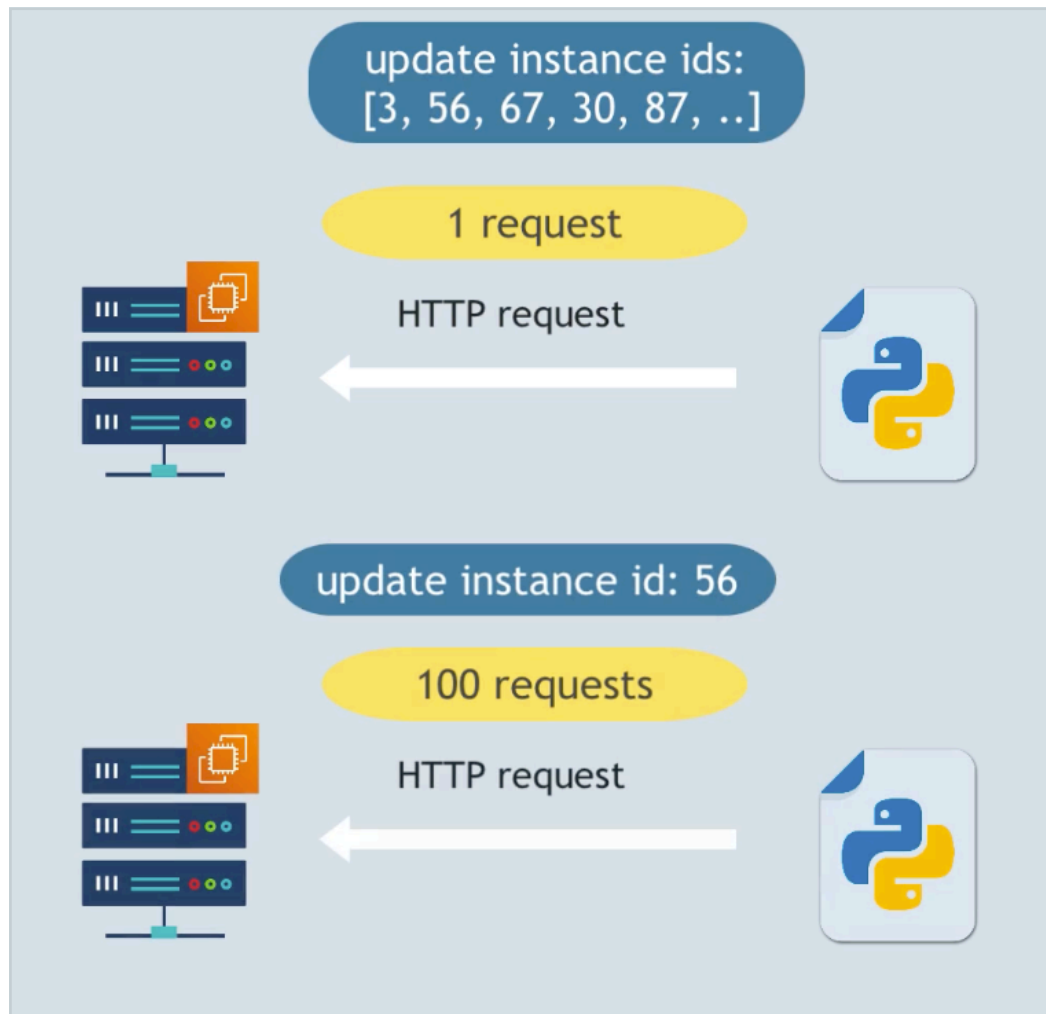
Now collect the Ids into a list and then tag instead of getting one instance at the time and tagging each one at the time

1) Collect all Instance Ids into a list

2) Add Tags for all InstanceIds at once

Instead of doing an AWS call for each Instance

Making one request to update multiple servers is more efficient than making multiple request and update one server at the time:

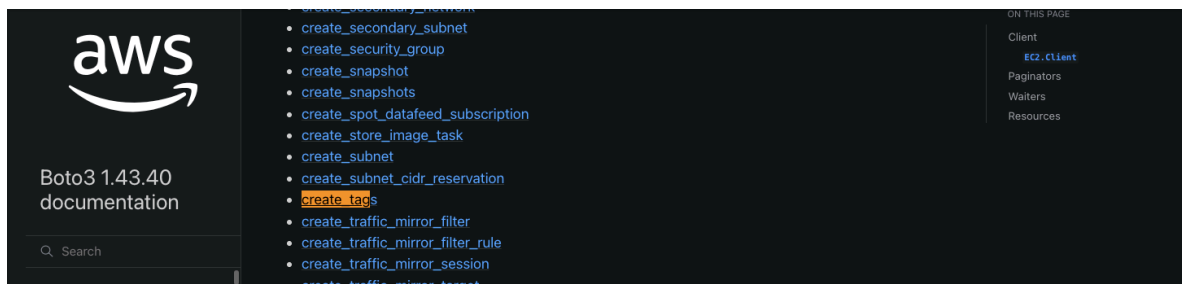


For that we need to have a global list to store the ids when iterating (see lines 5 and 11)

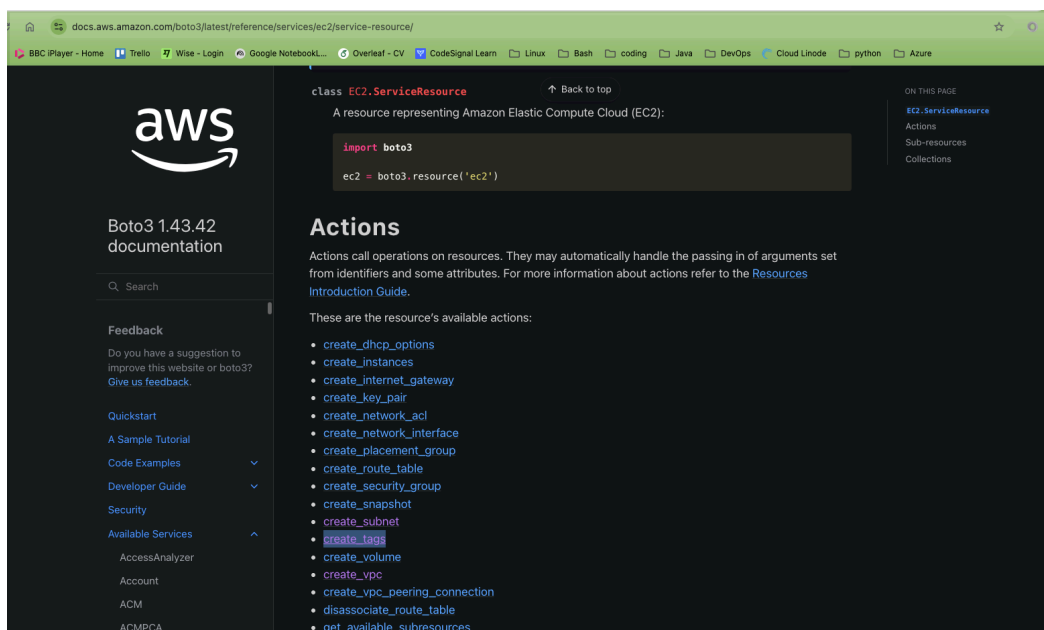
```
add-env-tags.py x
1 import boto3
2
3 ec2_client_frankfurt = boto3.client(*args: 'ec2', region_name='eu-central-1')
4
5 instances_ids_frankfurt = [] You, 4 minutes ago • Uncommitted changes
6
7 reservations_frankfurt = ec2_client_frankfurt.describe_instances()['Reservations']
8 for res in reservations_frankfurt:
9     instances = res['Instances']
10     for ins in instances:
11         instances_ids_frankfurt.append(ins['InstanceId'])
```

Now let's add the environment tag to the instances found in the list:
And we use the function `.create_tags()`

which can be found in the client



And also in the resource



So we will use the resource! We will execute `create_tags` on the EC2 resource

ServiceResource / Action / create_tags

ON THIS PAGE
EC2.ServiceResource.create_tags
{}

create_tags

EC2.ServiceResource.create_tags(kwargs)**

Adds or overwrites only the specified tags for the specified Amazon EC2 resource or resources. When you specify an existing tag key, the value is overwritten with the new value. Each resource can have a maximum of 50 tags. Each tag consists of a key and optional value. Tag keys must be unique per resource.

For more information about tags, see [Tag your Amazon EC2 resources](#) in the *Amazon Elastic Compute Cloud User Guide*. For more information about creating IAM policies that control users' access to resources based on tags, see [Supported resource-level permissions for Amazon EC2 API actions](#) in the *Amazon Elastic Compute Cloud User Guide*.

See also: [AWS API Documentation](#)

Request Syntax

```
response = ec2.create_tags(
    DryRun=True|False,
    Resources=[
        'string',
    ],
    Tags=[
        {
            'Key': 'string',
            'Value': 'string'
        },
    ]
)
```

In the 'Resources' block we will pass the Ids of the resources.

So we need to create a resource

```
add-env-tags.py x
1 import boto3
2
3 ec2_client_frankfurt = boto3.client(*args: 'ec2', region_name='eu-central-1')
4 ec2_resource_frankfurt = boto3.resource(*args: 'ec2', region_name='eu-central-1') You, A minute ago
5
6 instances_ids_frankfurt = []
7
```

and then we make the call to create_tags()

```
14 response = ec2_resource_frankfurt.create_tags(
15     Resources=instances_ids_frankfurt,
16     Tags=[
17         {
18             'Key': 'environment',
19             'Value': 'prod'
20         },
21     ]
22 ) You, A minute ago • Uncommitted changes
```

The full code so far is:

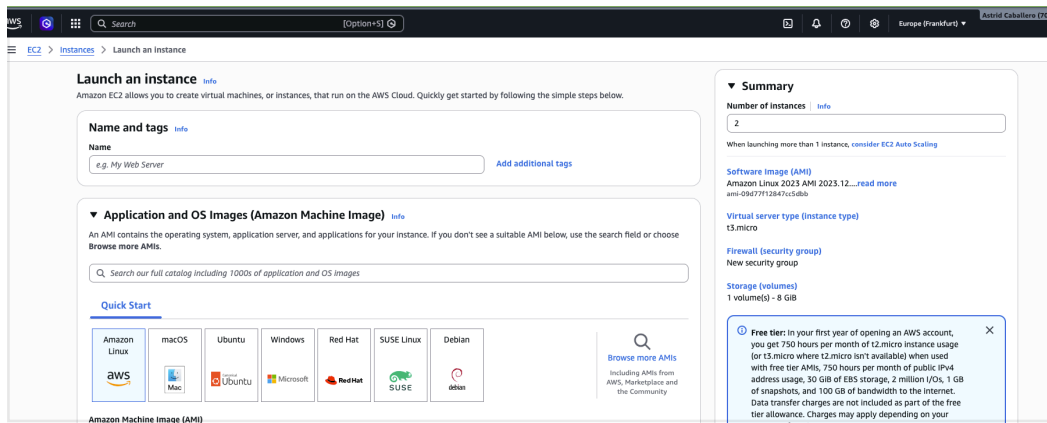
```
add-env-tags.py x
1 import boto3
2
3 ec2_client_frankfurt = boto3.client(*args: 'ec2', region_name='eu-central-1')
4 ec2_resource_frankfurt = boto3.resource(*args: 'ec2', region_name='eu-central-1')
5
6 instances_ids_frankfurt = []
7
8 reservations_frankfurt = ec2_client_frankfurt.describe_instances()['Reservations']
9 for res in reservations_frankfurt:
10     instances = res['Instances']
11     for ins in instances:
12         instances_ids_frankfurt.append(ins['InstanceId'])
13
14 response = ec2_resource_frankfurt.create_tags(
15     Resources=instances_ids_frankfurt,
16     Tags=[
17         {
18             'Key': 'environment',
19             'Value': 'prod'
20         },
21     ]
22 )
You, A minute ago • Uncommitted changes
```

And now we do the same for Paris

```
6 ec2_client_Paris = boto3.client(*args: 'ec2', region_name='eu-west-3')
7 ec2_resource_Paris = boto3.resource(*args: 'ec2', region_name='eu-west-3')
8
9 instances_ids_frankfurt = []
10 instances_ids_Paris = []
...
28 reservations_Paris = ec2_client_Paris.describe_instances()['Reservations']
29 for res in reservations_Paris:
30     instances = res['Instances']
31     for ins in instances:
32         instances_ids_Paris.append(ins['InstanceId'])
33
34 response = ec2_resource_Paris.create_tags(
35     Resources=instances_ids_Paris,
36     Tags=[
37         {
38             'Key': 'environment',
39             'Value': 'dev'
40         },
41     ]
42 )
You, Moments ago • Uncommitted changes
```

To test this python script we will create some servers in AWS manually without TF just for the DEMO

So two in Frankfurt



Instances (2) [Info](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#)

Saved filter sets: Choose filter set [Find Instance by attribute or tag \(case-sensitive\)](#)

[Instance state = running](#) [Clear filters](#)

☐	Name ↗	Instance ID	Instance state ↕	Instance type ↕	Status check ↕	Availability Zone ↕	Public IPv4 DNS ↕	Public IPv4 ... ↕	Elastic IP
☐		i-089eec74cf9c3d68	● Running 🔍 🔗	t3.micro	🔄 Initializing	eu-central-1c	ec2-3-72-81-115.eu-ce...	3.72.81.115	-
☐		i-0bbecaa852a06477a	● Running 🔍 🔗	t3.micro	🔄 Initializing	eu-central-1c	ec2-3-78-93-33.eu-cent...	3.78.93.33	-

Now let's launch 1 server in Paris

Instances (1) [Info](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets: Choose filter set [Find Instance by attribute or tag \(case-sensitive\)](#)

[Instance state = running](#) [Clear filters](#)

☐	Name ↗	Instance ID	Instance state ↕	Instance type ↕	Status check ↕	Availability Zone ↕	Public IPv4 DNS ↕	Public IPv4 ... ↕	Elastic IP	IPv6 IPs
☐		i-015f96f59fd8e6f05	● Running 🔍 🔗	t3.micro	🔄 Initializing	eu-west-3c	ec2-13-39-106-102.eu-...	13.39.106.102	-	-

And the servers don't have tags yet, see example for the Paris server:

Instances > i-015f96f59fd8e6f05

IPv6 address: -

Hostname type: IP name: ip-172-31-45-160.eu-west-3.compute.internal

Answer private resource DNS name: IPv4 (A)

Auto-assigned IP address: 13.39.106.102 [Public IP]

IAM role: -

IMDSv2: Required

Operator: -

Instance state: ● Running

Private IP DNS name (IPv4 only): ip-172-31-45-160.eu-west-3.compute.internal

Instance type: t3.micro

VPC ID: vpc-0424e806da8ab5368

Subnet ID: subnet-0a2502d07b395d48a

Instance ARN: arn:aws:ec2:eu-west-3:706976333940:instance/i-015f96f59fd8e6f05

Public DNS: ec2-13-39-106-102.eu-west-3.compute.amazonaws.com

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations

Auto Scaling Group name: -

Managed: false

Tags

Key: Value

No tags found

[Manage tags](#)

And now we execute the python script


```
Run add-env-tags x
/Users/astrid/PythonProject/Automation_with_python/.venv/bin/python
Process finished with exit code 0
```

And check AWS UI for Frankfurt region and we see that both servers have the environment tag with value 'prod'

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
☒	i-089eec474cf9c3d68	Running	t3.micro	3/3 checks passed	eu-central-1c	ec2-3-72-81-115.eu-ce...	3.72.81.115	-
☐	i-0bbecaa852a06477a	Running	t3.micro	3/3 checks passed	eu-central-1c	ec2-3-78-93-33.eu-cent...	3.78.93.33	-

i-089eec474cf9c3d68

Details Status and alarms Monitoring Security Networking Storage Tags

Tags

environment prod

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	i-089eec474cf9c3d68	Running	t3.micro	3/3 checks passed	eu-central-1c	ec2-3-72-81-115.eu-ce..
☒	i-0bbecaa852a06477a	Running	t3.micro	3/3 checks passed	eu-central-1c	ec2-3-78-93-33.eu-cent

i-0bbecaa852a06477a

Details Status and alarms Monitoring Security Networking Storage Tags

Tags

environment prod

And Paris has the tag as well with value 'dev'

ch

[Option+S]

Europe (Paris)

Alt

Instances (1/1) info

Last updated less than a minute ago

Connect

Instance state

Actions

Saved filter sets

Choose filter set

Find instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Name

Instance ID

Instance state

Instance type

Status check

Availability Zone

Public IPv4 DNS

Public IPv4 ...

Elastic IP

i-015f96f59fd8e6f05

Running

t3.micro

3/3 checks passed

eu-west-3c

ec2-13-39-106-102.eu-...

13.39.106.102

-

i-015f96f59fd8e6f05

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Tags

Key

Value

environment

dev